

Changin Oh

PhD Candidate
Department of Mathematics
Toronto Metropolitan University

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Education

Toronto Metropolitan University

2023 – Present

PhD in Mathematics

Toronto, ON

- Supervisor: Kathleen Wilkie
- Thesis: “Reliable Mathematical Modelling in Oncology: Model Formulation, Parameter Estimation, and Data-Driven Discovery”

York University

2021 – 2023

MA in Mathematics

North York, ON

- Supervisor: Iain Moyles
- Thesis: “Linear Spectral Unmixing Algorithms for Abundance Fraction Estimation in Spectroscopy”

University of Western Ontario

2009 – 2017

BSc in Mathematics

London, ON

- Minor in Japanese Studies
- Dean’s Honour List for 2009 – 2011

Publications

Peer-Reviewed Journal Articles

1. M. Oh, **C. Oh**, T. Ghosh, and E. Tabassum, “Selective effects of backbone cyclization and disulfide bonding as global covalent constraints on the conformational ensemble of sunflower trypsin inhibitor-1,” *The Journal of Physical Chemistry B*, vol. 130, no. 28, pp. 7075–7084, Jul 2026.
2. **C. Oh** and K. P. Wilkie, “How mathematical forms of chemotherapy and radiotherapy bias model-optimized predictions: Implications for model selection,” *Mathematical Biosciences*, vol. 396, p. 109685, Jun 2026.
Featured as a research highlight in *This Week in Mathematical Oncology*, Issue 370.
3. M. Oh, **C. Oh**, and E. Tabassum, “Covalent: Interpretable and discriminative collective variables reveal ligand-dependent switching in human cellular retinol-binding protein 2,” *Journal of Chemical Theory and Computation*, vol. 21, no. 23, pp. 12356–12370, Dec 2025.
4. M. Oh, **C. Oh**, and D. F. Weaver, “Effect of cholesterol on the structure of networked water at the surface of a model lipid membrane,” *The Journal of Physical Chemistry B*, vol. 124, no. 18, pp. 3686–3694, May 2020.
5. M. Oh, **C. Oh**, M. Gupta, and D. F. Weaver, “Decoding interfacial water orientation to predict surface charge density on a model sheet using a deep learning algorithm,” *The Journal of Physical Chemistry C*, vol. 124, no. 4, pp. 2574–2582, Jan 2020.

6. M. Oh, M. Gupta, **C. Oh**, and D. F. Weaver, “Understanding the effect of nanoconfinement on the structure of water hydrogen bond networks,” *Physical Chemistry Chemical Physics*, vol. 21, no. 47, pp. 26237–26250, Dec 2019.

Blog Posts

1. A. Köhn-Luque, J. Nardini, **C. Oh**, L. Podina, G. Powathil, E. Rutter, J. West, and K. Wilkie, “Learning dynamics with neural networks,” *Mathematical Oncology Blog*, Mar 2025.

Presentations

Toroidal Search Algorithm: A Topology-Inspired Metaheuristic

- 2026 Complex Systems Day (Toronto) 2026/08
- Mathematical Biology Seminar (Toronto Metropolitan University) 2026/02

A Gentle Introduction to Riemannian Gradient Descent Method on Smooth Manifolds

- Chemistry Research Group (Memorial University of Newfoundland) 2026/01
- Mathematical Biology Seminar (Toronto Metropolitan University) 2025/10

Aging Effects in Cancer: An Age-Dependent Tumour-Immune Model

- 2025 Complex Systems Day (Toronto) 2025/08
- Mathematical Biology Seminar (Toronto Metropolitan University) 2024/02

How Mathematical Forms Describing Cytotoxic Action of Chemotherapy and Radiotherapy Bias Predictions

- Workshop on Mechanistic Learning as a Combination of Machine Learning and Modelling in Mathematical Oncology (Poster – Banff International Research Station) 2025/01
- Workshop on Mathematical Modelling of Cancer Treatments, Resistance, Optimization (Poster – Fields Institute) 2024/09

Teaching-Learning-Based Optimization with Applications in Analytic Chemistry

- Mathematical Biology Seminar (Toronto Metropolitan University) 2023/09

Optimization Techniques for Linear Spectral Unmixing in Spectroscopy

- Mathematics & Statistics Seminar (York University) 2022/10

Awards and Prizes

Ontario Graduate Scholarship (\$15,000 each) 2024 – 2025, 2026 – 2027

Prestigious, competitive provincial scholarship awarded to graduate students in Ontario.

Toronto Metropolitan Graduate Fellowship (Total: \$16,250) 2023 – 2024, 2025 – 2026

Competitive, merit-based institutional fellowship for graduate students demonstrating academic excellence.

Mathematics Graduate Award (Total: \$11,650) 2023 – 2026

Departmental award recognizing outstanding academic performance in graduate mathematics.

Toronto Metropolitan Graduate Student Travel Fund Award (\$500) 2025

Institutional support for research dissemination and active participation at academic conferences.

Toronto Metropolitan Graduate Scholarship (declined) (\$15,000)	2024 – 2025
Competitive institutional scholarship awarded through the Ontario Graduate Scholarship competition.	
Toronto Metropolitan Graduate Development Award (\$2,550)	2024 – 2025
Departmental award recognizing academic potential and professional success.	
York Graduate Fellowship (\$5,403)	2021 – 2022
Institutional fellowship supporting academic and research activities.	
Classical Studies Prize in Elementary Latin (\$150)	2016
Awarded to the student with the highest mark in Latin 1000.	
Class of '49 Prize (\$150)	2010
Awarded to the student with the highest mark in Calculus 1501.	
Western Scholarship of Excellence (\$8,000)	2009
Merit-based entrance scholarship recognizing outstanding high school academic achievement.	

Honours and Distinctions

Featured Research Article	2026/05
<i>Mathematical Oncology Newsletter</i>	
▫ The article “How mathematical forms of chemotherapy and radiotherapy bias model-optimized predictions: Implications for model selection” was featured in <i>This Week in Mathematical Oncology</i>, Issue 370.	

Research Experience

Collaborating Researcher	2025/09 – Present
<i>Memorial University of Newfoundland</i>	
<i>St. John's, NL</i>	
▫ Implemented and enhanced a Riemannian gradient descent algorithm for multiclass harmonic linear discriminant analysis on the Stiefel manifold. ▫ Formulated a novel global twist angle for cyclic peptides and developed its computational implementation. ▫ Developed a Grassmann manifold framework for the mathematical analysis and comparison of multidimensional collective variables.	
Collaborating Researcher	2019/07 – 2020/06
<i>Krembil Research Institute</i>	
<i>Toronto, ON</i>	
▫ Conducted topological analyses of hydrogen bond networks in confined and interfacial water. ▫ Implemented a deep learning algorithm for indirect surface characterization. ▫ Investigated the effects of cholesterol on water structure at the surface of a model lipid membrane.	
Volunteer Research Assistant	2015/05 – 2015/08
<i>University of Western Ontario</i>	
<i>London, ON</i>	
▫ Investigated nanosized aqueous droplets using molecular dynamics simulations.	

Teaching Experience

Teaching Assistant

2023/09 – 2027/04

Toronto Metropolitan University

Toronto, ON

- Led weekly labs guiding students through difficult problems and reviewing course material.
- Contributed to the development of exam marking schemes (MTH 231).

MTH 141 – Linear Algebra

MTH 207 – Calculus and Computational Methods I

MTH 231 – Modern Mathematics II

MTH 240 – Calculus II

MTH 330 – Calculus and Geometry

MTH 350 – Multivariate and Vector Calculus

MTH 501/510 – Numerical Analysis

MTH 430 – Dynamic Systems Differential Equations

MTH 719 – Applied Linear Algebra

Guest Instructor

2026/01 – 2026/05

Memorial University of Newfoundland

St. John's, NL

- Delivered intuition-focused lectures in general topology for a chemistry research group.

Teaching Assistant

2021/09 – 2022/04

York University

North York, ON

- Provided one-on-one and group tutoring to students.

MATH 1013 – Applied Calculus I

MATH 1025 – Applied Linear Algebra

MATH 1300 – Differential Calculus with Applications

MATH 1507 – Mathematics for the Life Sciences

Teaching Assistant

2013/09 – 2014/04

University of Western Ontario

London, ON

- Led group conversation tutorials to support students in improving their Japanese speaking skills.

JAPANESE 1036 – Japanese for Beginners

JAPANESE 2260 – Intermediate Japanese

Educational Outreach

Instructor

2026/02 – 2026/06

Toronto Garden Church

North York, ON

- Taught mathematics to students in Grades 7–10 as part of a community outreach program.

Instructor

2025/03 – 2025/06

Toronto Milal Church

Toronto, ON

- Taught essential computer skills to elderly learners, including Microsoft Word, Paint, and introductory AI tools.

Instructor

2021/08 – 2024/07

Online Learning Community

Korea

- Taught linear algebra, calculus, probability theory, and real analysis online to learners without strong mathematical backgrounds.

Instructor

2015/09 – 2016/08

London Full Gospel Church

London, ON

- Taught mathematics and Japanese to secondary school students as part of a community outreach program.

Instructor

2014/09 – 2015/08

*Oakridge Presbyterian Church**London, ON*

- Taught mathematics to secondary school students as part of a community outreach program.

Instructor

2008/05 – 2009/07

*Oakville Presbyterian Church**Oakville, ON*

- Taught mathematics and Japanese to secondary school students as part of a community outreach program.

Technical and Language Skills

- Julia, MATLAB, Python
- Korean (native), English (advanced), Japanese (advanced, JLPT N1)